Another common notation for performing computations involving Hamiltonian systems is the *Poisson bracket* which is defined for two smooth scalar-valued functions g_1 and g_2 of the phase variables $(\boldsymbol{q},\boldsymbol{p})$ of a Hamiltonian system in $\mathbb{R}^m$ by

$$\{g_1, g_2\} = \sum_{i=1}^{\mathrm{N_c}} \left(\frac{\partial g_1}{\partial q_i}\frac{\partial g_2}{\partial p_i} - \frac{\partial g_2}{\partial q_i}\frac{\partial g_1}{\partial p_i} \right) = \nabla g_1^T \boldsymbol{J} \nabla g_2,$$

where $\boldsymbol{J}$ is the skew symmetric symplectic structure matrix:

$$\boldsymbol{J} = \begin{bmatrix} 0 & \boldsymbol{I} \\ -\boldsymbol{I} & 0 \end{bmatrix},$$

and $\boldsymbol{I}$ represents the $\mathrm{N_c} \times \mathrm{N_c}$ identity matrix. The Poisson bracket has the following properties, which are easily verified from the definition (for g_1, g_2, g_3 being three arbitrary functions of the phase variables):

Bilinearity $\{g_1, \alpha g_2 + \beta g_3\} = \alpha\{g_1, g_2\} + \beta\{g_1, g_3\}$.
Skew symmetry $\{g_1, g_2\} = -\{g_2, g_1\}$ and this implies $\{g_1, g_1\} = 0$.
Jacobi identity $\{g_1, \{g_2, g_3\}\} + \{g_3, \{g_1, g_2\}\} + \{g_2, \{g_3, g_1\}\} = 0$.

In terms of the Poisson bracket, it is possible to write the differential equation corresponding to a coordinate q_i, say, as

$$\dot{q}_i = \{q_i, H\}.$$

More generally, if $F(\boldsymbol{q},\boldsymbol{p})$ is any smooth, scalar-valued function of the phase variables, we may write

$$\dot{F} = \frac{\mathrm{d}}{\mathrm{d}t} F(\boldsymbol{q}(t),\boldsymbol{p}(t)) = \{F, H\}.$$

As a consequence, we have the following relation between the Lie derivative and the Poisson bracket:

$$\mathcal{L}_{\boldsymbol{J}\nabla H} F \equiv \{F, H\}.$$

For a Hamiltonian flow, we typically simplify notation by writing $\mathcal{L}_H$ in place of $\mathcal{L}_{\boldsymbol{J}\nabla H}$.